

Lab 3 Activity

PSYC 6802: Intro to Psychological Statistics

In the Lab 3 slides, we loaded in and manipulated the `Airlines.csv` data. Run all of these commands first.

```
library(tidyverse)

flights <- read_csv("Airlines.csv")

flights <- flights %>%
  mutate(Delay = factor(Delay,
                        levels = c("0", "1"), # values observed in the data
                        labels = c("On Time", "Delay"))) # meaningful labels
```

1

Use the χ^2 test of independence to test whether there is an association between whether a flights gets delayed depending on whether it is operated by Delta Airlines (DL) or Southwest Airlines (WN). Do this only for flights originating from Los Angeles (LAX) and flying to Salt Lake City (SLC).

- Step 1. Use `filter` to retain only those flights that we want.
- Step 2. Create a 2x2 contingency table showing the number of flights by airline and delay status.
- Step 3. Run a χ^2 test of independence with the continuity correction.

2

Compute the W coefficient for the test in #1.

3

Write $\chi^2(1, N = 242) = 9.71$, $p = .002$, $W = .2$ using Quarto or R Markdown (the outputted document should render the mathematical notation properly).

4

Run Fisher's exact test on these data. Which test do you think is more appropriate?

5

Based on your results, should you choose Delta or Southwest if you were to fly from Los Angeles to Salt Lake city? Feel free to calculate other types of effects to inform your answer.